Understanding Diabetes Progression

A reproducible analysis of patient characteristics and one-year progression

Scenario 1 | Data Analytics and Statistics with Applications

Research question. Which patient characteristics are associated with diabetes progression, what do regression models tell us, and what conclusions can reasonably be drawn?

Executive summary. In this observational dataset of 442 patients, BMI, the serum measure s5 and blood pressure showed the strongest positive linear associations with the one-year progression measure. A multiple model using these three variables explained 48.0% of observed variation (adjusted $R^2 = 0.477$). The results describe associations in this dataset; they do not establish that changing any predictor would cause a change in progression.

1. Data and preparation

The supplied dataset contains 442 rows and 11 columns. Each row represents one patient. The outcome is a continuous measure of diabetes progression one year after baseline; the predictors are age, sex, BMI, blood pressure and six serum measurements (s1-s6). The sex field is numerically coded but represents categories, so it was treated as a factor and excluded from Pearson correlations among numerical measurements.

Data checks found no missing values and no duplicated rows. The supplied file contains plausible ranges for its recorded variables; this check does not validate measurements against medical records. Health data require privacy and responsible handling. These data are de-identified in the supplied teaching file, but findings should not be used to make individual clinical decisions.

Source: NCSU-hosted diabetes dataset supplied with the project brief. The analysis is for coursework and is not clinical guidance.

2. Exploratory analysis

The progression measure averaged 152.1 (SD 77.1; median 140.5; range 25-346). BMI averaged 26.4 (SD 4.4; median 25.7; range 18.0-42.2). The outcome varies substantially between patients, supporting the use of a model that describes average trends while recognizing considerable individual-level variation.

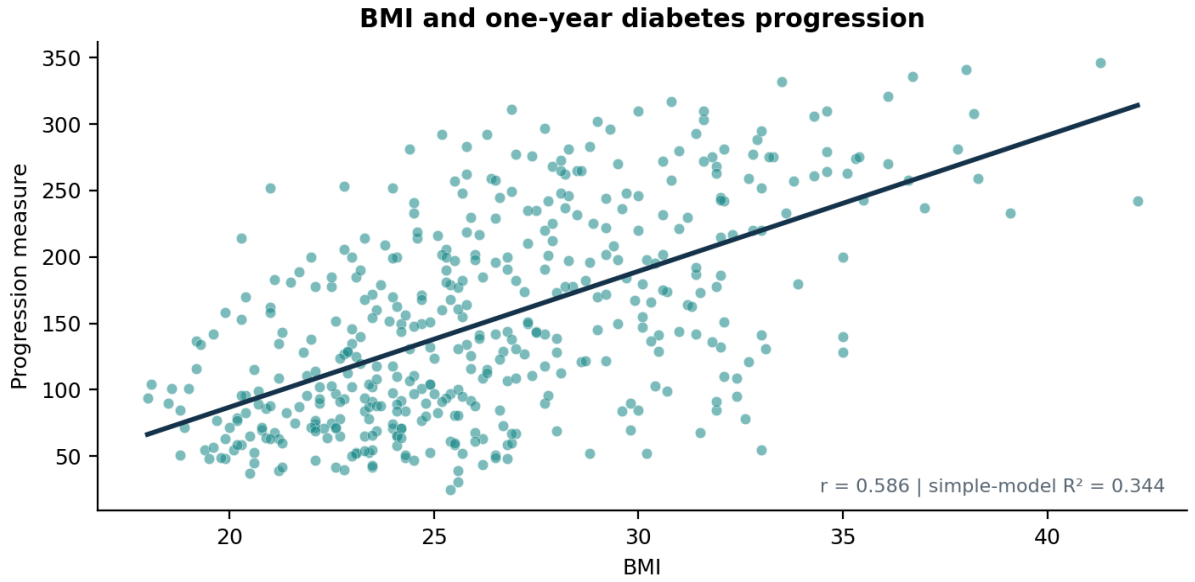


Figure 1. BMI and the progression measure. The fitted line summarizes the average linear association; the spread around it shows that BMI alone does not predict each patient's outcome precisely.

The outcome is broadly unimodal with a longer upper tail and several high values. BMI is more concentrated and approximately unimodal, with a few observations at higher values. These observations are retained; there is no project-supported reason to remove them. Descriptive patterns do not by themselves establish a model or identify causes.

Key descriptive message

There is substantial between-patient variation in progression, while BMI covers a moderate range. The data are suitable for exploratory association analysis, but prediction uncertainty remains important.

3. Relationships between variables

Pearson correlations identify BMI (r = 0.586), s5 (r = 0.566) and blood pressure (bp; r = 0.441) as the strongest positive linear associations with progression. The serum measure s3 is negatively associated (r = -0.395). Age (r = 0.188) and sex (not included in the Pearson matrix) are not treated as strong numerical linear signals here. Correlation is a screening description, not evidence of an independent effect.

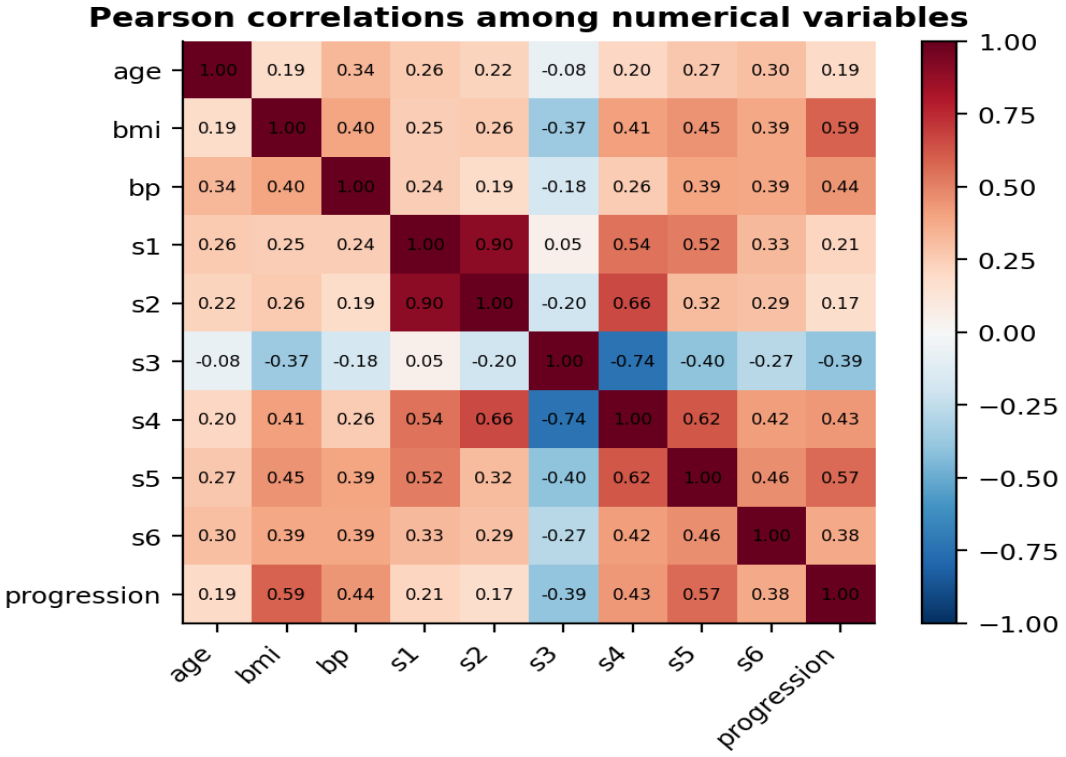


Figure 2. Pearson correlation matrix for numerical variables. Sex is omitted because it is categorical. Correlated predictors may share information; the selected model's VIF values are low (approximately 1.27-1.34), so there is no indication of serious multicollinearity among BMI, bp and s5 in this model.

BMI and s5 are both promising for further analysis, and bp adds a third clinically interpretable measurement. The model uses a small, exploratory predictor set selected from observed associations. The choice is not a claim that these are the only relevant predictors.

4. Regression models

The simple linear regression using BMI was: **progression = -117.77 + 10.23 × BMI**. On average, a one-unit higher BMI was associated with 10.23 higher progression-measure units in this sample (95% CI 8.91 to 11.56; $p < 0.001$). Its R^2 was 0.344, meaning BMI alone accounted for about 34.4% of the observed variation in this linear model. The intercept is an extrapolation at BMI = 0 and has no useful patient-level interpretation.

The multiple regression using BMI, bp and s5 was: **progression = -334.88 + 6.50 × BMI + 0.90 × bp + 49.58 × s5**. Holding the other two predictors constant, a one-unit higher BMI was associated with 6.50 higher progression units (95% CI 5.13 to 7.87; $p < 0.001$). Holding BMI and s5 constant, one unit higher bp was associated with 0.90 higher progression units (95% CI 0.48 to 1.33; $p < 0.001$). For s5, the corresponding estimate was 49.58 (95% CI 38.00 to 61.15; $p < 0.001$).

Model	R^2	Adjusted R^2	Residual SD
BMI only	0.344	0.342	62.5
BMI + bp + s5	0.480	0.477	55.8

The multiple model raised R^2 to 0.480 and reduced residual SD from 62.5 to 55.8. That is a better in-sample fit, but it does not show that the model is clinically useful for individual predictions or that it will perform similarly in another population.

5. Model evaluation and uncertainty

The residuals-versus-fitted plot (generated by the R script) should be reviewed for curvature, changing spread and extreme residuals. The independent numerical check shows residuals distributed on both sides of zero across fitted-value quartiles, with wider residual spread in some middle-to-upper fitted ranges; this is a caution that a simple linear model does not capture all outcome variation. Formal predictive validation was not part of this assignment.

The selected model's VIF values are low (about 1.27 for bp, 1.34 for BMI and 1.34 for s5), which suggests that severe linear redundancy among these three predictors is unlikely. VIF does not assess omitted-variable bias, measurement error, nonlinear relationships or causality.

The coefficient confidence intervals quantify uncertainty under the model assumptions. For example, the BMI association in the multiple model is estimated at 6.50 units per BMI unit, with a 95% interval from 5.13 to 7.87. Statistical significance indicates evidence against a zero linear coefficient under the model; practical importance depends on the measurement scale, clinical context and whether the association is causal, none of which is established by this analysis.

6. Conclusions, limitations and responsible interpretation

This is observational data, not a randomized experiment. Regression coefficients therefore describe conditional associations and should not be interpreted as causal effects. Potential confounders, treatment history, disease severity at baseline, lifestyle factors and other clinical details are not represented in the supplied analysis. Predictor measurement and the construction of the progression score may also affect interpretation.

The results apply directly only to the patients represented in this dataset. Generalization to other populations, settings or time periods would require external evidence. The supported conclusion is that BMI, bp and s5 are associated with the one-year progression measure in this sample, and the three-variable model describes more of its variation than BMI alone. The analysis does not show that changing BMI, blood pressure or s5 will change progression, nor does it establish a clinical decision rule.

Reproducibility and references

The accompanying R script reproduces data checks, descriptive statistics, plots, the correlation matrix, both regressions, coefficient confidence intervals, residual diagnostics and VIFs. Data source: NCSU-hosted diabetes dataset, https://www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt. Project requirements and scenario description: supplied Final Project Portfolio Instructions and Scenario 1 Diabetes Brief.